Software Requirements Specification (SRS)

LokVeda — Digital E-Gram Panchayat System

Version: 2.0.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of LokVeda — Digital E-Gram Panchayat System.

LokVeda is a web-based governance platform developed to modernize village-level administrative processes through secure digital service delivery, workflow management, and citizen-centric interactions.

The system provides:

- Aadhaar-based authentication
- OTP verification
- Location-aware access control
- Role-based authorization
- Service application management
- Administrative approval workflows
- Panchayat-level service management

This document serves as a reference for development, testing, deployment, maintenance, and academic evaluation.

1.2 Scope

LokVeda is designed as a secure digital governance solution for Gram Panchayats.

The platform aims to reduce manual administrative processes by providing a centralized system through which citizens, staff members, and administrators can interact digitally.

The current implementation includes:

Citizen Services

- Secure login
- Service application submission
- Application tracking
- Re-submission of rejected applications
- Profile viewing

Staff Operations

- Application review
- Application rejection
- Verification workflow management

Administrative Operations

- Service activation and deactivation
- Application approval
- Rejected application review
- Approved application management

Security Features

- OTP authentication
- JWT session management
- Geographic access verification
- Account protection
- Suspicious activity detection

The platform is intended for educational, demonstration, and future deployment purposes.

1.3 Intended Audience

This document is intended for:

Academic Evaluators

To understand system objectives, architecture, and implementation.

Developers

To understand requirements, workflows, and system design.

Maintainers

To support future enhancements and maintenance activities.

Project Stakeholders

To understand the capabilities and limitations of the system.

1.4 Definitions, Acronyms and Abbreviations

Term	Description
Aadhaar	Government-issued identification number
OTP	One Time Password
JWT	JSON Web Token
MVC	Model View Controller
UI	User Interface
UX	User Experience
SRS	Software Requirements Specification
Admin	Panchayat Administrator
Staff	Panchayat Employee
Citizen	Registered User
API	Application Programming Interface
DB	Database
RBAC	Role Based Access Control

1.5 References

The following resources were used during the design and development of LokVeda:

1. IEEE Software Requirements Specification Standards
2. MongoDB Documentation
3. Mongoose Documentation
4. Express.js Documentation
5. Node.js Documentation
6. MDN Web Docs
7. JSON Web Token Documentation
8. bcrypt Documentation
9. Nodemailer Documentation
10. Government E-Governance Service Models
11. OpenAI Documentation Assistance

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2. Overall Description

2.1 Product Perspective

LokVeda is a standalone web application following the MVC architectural pattern.

The system consists of:

Frontend Layer

Responsible for:

- User interaction
- Form rendering
- Navigation
- Visual presentation

Technologies:

- HTML
- CSS
- JavaScript
- EJS

Backend Layer

Responsible for:

- Authentication
- Authorization
- Business logic
- Workflow management
- Validation

Technologies:

- Node.js
- Express.js

Database Layer

Responsible for:

- User management
- Service management
- Area information
- Application storage

Technologies:

- MongoDB
- Mongoose

Security Layer

Responsible for:

- OTP verification
- Session validation
- Role enforcement
- Geographic verification
- Login protection

Technologies:

- JWT
- bcrypt
- Crypto
- Validator

Deployment Layer

Responsible for:

- Application hosting and deployment platform used for running the Node.js and Express.js server in a cloud environment

Technologies:

- Render

2.2 Product Functions

LokVeda provides the following primary functions:

Authentication

- Aadhaar verification
- OTP generation
- OTP validation
- Session creation
- Session termination

Authorization

- Citizen access control
- Staff access control
- Administrator access control

Service Management

- Service activation
- Service deactivation
- Service visibility management

Application Management

- Application submission
- Application review
- Application approval
- Application rejection
- Application tracking

Profile Management

- Citizen profile viewing
- Staff profile viewing
- Administrative profile viewing

Monitoring

- Login tracking
- Session tracking
- Audit information tracking

2.3 User Classes and Characteristics

Citizen

Characteristics:

- Non-technical users
- Village residents
- Service applicants

Capabilities:

- Submit applications
- View application status
- Re-submit rejected applications

Staff

Characteristics:

- Panchayat employees
- Verification authority

Capabilities:

- Review applications
- Reject applications
- Forward applications for approval

Administrator

Characteristics:

- Panchayat decision makers

Capabilities:

- Approve applications
- Manage services
- Review rejected applications
- Monitor service availability

2.4 Operating Environment

LokVeda operates within the following environment:

Client Side

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Mobile Browsers

Server Side

- Node.js Runtime Environment

Database

- MongoDB

Network

- Internet Connection
- HTTPS Communication

2.5 Design Constraints

The system is constrained by:

- Browser-based execution
- Internet dependency
- Geographic verification requirements
- OTP delivery dependency
- Database connectivity requirements

2.6 Assumptions and Dependencies

The system assumes:

- Users are pre-registered
- Aadhaar records are valid
- Email addresses are available
- Internet connectivity exists
- Location services are available
- MongoDB database remains accessible

Failure of these assumptions may limit functionality.

3. System Features

3.1 Authentication System

Description

The Authentication System is responsible for verifying user identity before granting access to protected resources.

LokVeda follows a pre-registration model where users are enrolled by the Panchayat administration and cannot independently create accounts.

Authentication is performed using:

- Aadhaar Number
- Geographic Verification
- One-Time Password (OTP)

Functional Requirements

AUTH-01

The system shall allow users to enter their Aadhaar number.

AUTH-02

The system shall verify whether the Aadhaar number exists in the database.

AUTH-03

The system shall verify whether the user account is approved.

AUTH-04

The system shall request geographic access permission before OTP generation.

AUTH-05

The system shall deny authentication when geographic verification fails.

AUTH-06

The system shall generate a One-Time Password after successful validation.

AUTH-07

The system shall send OTP to the registered email address.

AUTH-08

The system shall allow OTP verification.

AUTH-09

The system shall create a secure session after successful OTP verification.

AUTH-10

The system shall redirect authenticated users to their respective dashboard.

3.2 OTP Verification System

Description

The OTP Verification System provides an additional layer of security by validating user ownership of the registered email account.

Functional Requirements

OTP-01

The system shall generate OTPs using cryptographically secure methods.

OTP-02

The system shall store OTP values in hashed form.

OTP-03

The system shall never store OTP values in plain text.

OTP-04

The system shall verify OTPs using bcrypt comparison.

OTP-05

The system shall invalidate OTPs after successful verification.

OTP-06

The system shall limit OTP request frequency.

OTP-07

The system shall allow OTP resend requests only after cooldown expiration.

OTP-08

The system shall count invalid OTP submissions as failed login attempts.

3.3 Geographic Verification System

Description

The Geographic Verification System ensures users authenticate only from permitted locations.

Functional Requirements

GEO-01

The system shall request user location permission.

GEO-02

The system shall obtain user coordinates.

GEO-03

The system shall calculate distance from the permitted area.

GEO-04

The system shall verify citizen access within approved states.

GEO-05

The system shall verify staff access within a 10 km radius.

GEO-06

The system shall verify administrator access within a 250 km radius.

GEO-07

The system shall deny access when distance restrictions are violated.

GEO-08

The system shall display user-friendly feedback messages explaining violations.

3.4 Session Management System

Description

The Session Management System maintains authenticated user access while protecting against session abuse.

Functional Requirements

SES-01

The system shall issue authentication tokens during login.

SES-02

The system shall issue session tokens after successful OTP verification.

SES-03

The system shall remove temporary authentication tokens after successful verification.

SES-04

The system shall store session information securely.

SES-05

The system shall invalidate sessions during logout.

SES-06

The system shall prevent multiple active sessions for the same account.

SES-07

The system shall validate session integrity before protected route access.

SES-08

The system shall redirect unauthorized users to the authentication page.

3.5 Citizen Workflow

Description

Citizens interact with LokVeda to submit applications and track service requests.

Functional Requirements

CIT-01

Citizens shall view active services.

CIT-02

Citizens shall not view services that are unavailable in their area.

CIT-03

Citizens shall submit service applications.

CIT-04

Citizens shall view application status.

CIT-05

Citizens shall view approved applications.

CIT-06

Citizens shall view reviewed applications.

CIT-07

Citizens shall view pending applications.

CIT-08

Citizens shall view rejected applications.

CIT-09

Citizens shall re-submit rejected applications.

CIT-10

Citizens shall view personal profile information.

3.6 Staff Workflow

Description

Staff members verify applications before forwarding them to administrators.

Functional Requirements

STF-01

Staff shall access pending applications.

STF-02

Staff shall review submitted applications.

STF-03

Staff shall review applicant profiles.

STF-04

Staff shall reject incomplete applications.

STF-05

Staff shall forward reviewed applications to administrators.

STF-06

Staff shall access previously rejected applications.

STF-07

Staff shall record review information.

STF-08

Staff shall record review timestamps.

STF-09

Staff shall not approve applications.

3.7 Administrator Workflow

Description

Administrators perform final verification and approval of applications.

Functional Requirements

ADM-01

Administrators shall access reviewed applications.

ADM-02

Administrators shall approve reviewed applications.

ADM-03

Administrators shall reject reviewed applications.

ADM-04

Administrators shall access rejected applications.

ADM-05

Administrators shall review rejected applications.

ADM-06

Administrators shall view approved applications.

ADM-07

Administrators shall review applicant profiles.

ADM-08

Administrators shall record approval information.

ADM-09

Administrators shall record approval timestamps.

3.8 Service Management System

Description

Administrators manage service availability within Panchayat areas.

Functional Requirements

SRV-01

Administrators shall activate services.

SRV-02

Administrators shall deactivate services.

SRV-03

Service activation shall apply only to the assigned area.

SRV-04

Citizens shall only view active services.

SRV-05

Service changes shall not affect historical applications.

SRV-06

The system shall display active and inactive services separately.

3.9 Application Tracking System

Description

The Application Tracking System maintains application state and audit information.

Supported States

- Pending
- Reviewed
- Approved
- Rejected

Functional Requirements

APP-01

The system shall store application state.

APP-02

The system shall track reviewers.

APP-03

The system shall track approvers.

APP-04

The system shall track rejection authority.

APP-05

The system shall store review timestamps.

APP-06

The system shall store approval timestamps.

APP-07

The system shall store rejection timestamps.

APP-08

The system shall prevent invalid workflow transitions.

3.10 Greeting System

Description

The Greeting System provides personalized greetings.

Functional Requirements

GRT-01

The system shall display user names.

GRT-02

The system shall display time-aware greetings.

GRT-03

The system shall support sunrise and sunset calculations.

GRT-04

The system shall fall back to local time when geographic calculations fail.

3.11 Theme Management System

Description

The Theme Management System supports light and dark mode preferences.

Functional Requirements

THM-01

The system shall detect browser preferences.

THM-02

The system shall allow manual theme switching.

THM-03

The system shall store user preferences locally.

THM-04

The system shall restore previously selected preferences.

THM-05

The system shall apply themes consistently across all pages.

4. External Interface Requirements

4.1 User Interface Requirements

Overview

LokVeda is designed primarily for non-technical users residing in rural areas.

The interface emphasizes:

- Simplicity
- Accessibility
- Readability
- Minimal navigation complexity
- Mobile responsiveness

Authentication Interface

The authentication interface shall provide:

- Aadhaar number input
- OTP verification input
- Status message display
- Loading indicators
- Location permission guidance

The interface shall prevent duplicate submissions during active operations.

Dashboard Interface

The dashboard shall:

- Display role-specific actions
- Display personalized greetings
- Display navigation controls
- Maintain consistent visual hierarchy

Different dashboards shall be displayed for:

- Citizens
- Staff
- Administrators

Service Interface

The service interface shall:

- Display active services
- Display inactive services
- Display service descriptions
- Provide service-specific navigation

Services shall be visually separated using color-coded sections.

Application Interface

The application interface shall:

- Display applicant information
- Display application metadata
- Display workflow controls
- Display review information

The interface shall remain responsive on desktop and mobile devices.

Profile Interface

The profile interface shall display:

- Profile picture

- Name
- Aadhaar number
- Date of birth
- Contact information
- Location information
- User role

Theme Support

The user interface shall support:

- Light Mode
- Dark Mode

Theme selection shall persist between sessions.

4.2 Hardware Requirements

Client Requirements

Minimum:

- Smartphone or Computer
- Internet Connection
- Modern Browser
- Location Services Support

Recommended:

- Android Smartphone
- Desktop Computer
- Broadband Connection

Server Requirements

Minimum:

- Node.js Runtime Environment
- MongoDB Database Server

Recommended:

- Cloud Hosting Environment
- Managed MongoDB Deployment

4.3 Software Requirements

Development Environment

- Visual Studio Code
- Git
- GitHub
- Render

Frontend Technologies

- HTML5
- CSS3
- JavaScript
- EJS

Backend Technologies

- Node.js
- Express.js

Database Technologies

- MongoDB
- Mongoose

Security Libraries

- bcrypt
- jsonwebtoken
- crypto
- validator

Utility Libraries

- cookie-parser
- dotenv
- nodemailer

4.4 Communication Interfaces

LokVeda communicates using:

- HTTPS Requests
- Browser APIs
- Email Services

Communication channels include:

- Browser ↔ Server
- Server ↔ MongoDB
- Server ↔ Email Service

All authentication operations occur through secure server-side processing.

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5. Non-Functional Requirements

5.1 Security Requirements

Authentication Security

The system shall:

- Use OTP-based verification
- Use JWT-based sessions
- Hash sensitive verification data
- Validate session integrity

Access Control

The system shall:

- Enforce role-based access control
- Prevent unauthorized route access
- Restrict workflow actions by role

Geographic Security

The system shall:

- Verify user location
- Restrict access based on configured geographic rules

Account Protection

The system shall:

- Track failed login attempts
- Detect suspicious login patterns
- Apply temporary account restrictions

5.2 Performance Requirements

The system shall:

- Load pages within acceptable response times
- Render interfaces efficiently
- Minimize unnecessary database operations

Target Response Time:

- Authentication Operations: < 5 seconds
- Dashboard Rendering: < 2 seconds
- Service Loading: < 2 seconds

5.3 Reliability Requirements

The system shall:

- Recover gracefully from temporary failures
- Maintain workflow consistency
- Preserve application state integrity

The system shall avoid:

- Duplicate approvals
- Duplicate reviews
- Invalid workflow transitions

5.4 Availability Requirements

The system shall remain available whenever:

- Database connectivity exists
- Server infrastructure remains operational

Planned availability target:

99%+

5.5 Maintainability Requirements

The system shall:

- Follow modular architecture
- Follow MVC design principles
- Separate concerns across layers

The codebase shall support:

- Future feature additions
- Service expansion
- Workflow enhancement

5.6 Scalability Requirements

The system shall support:

- Additional services
- Additional Panchayat areas
- Additional users
- Additional administrative roles

Future scalability shall not require major architectural redesign.

5.7 Usability Requirements

The system shall:

- Provide user-friendly messages
- Use simple language
- Minimize technical terminology
- Maintain consistent navigation

The system shall remain usable by:

- First-time users
- Rural citizens
- Non-technical users

5.8 Accessibility Requirements

The system shall:

- Support keyboard navigation
- Maintain readable typography
- Provide adequate color contrast
- Support responsive layouts

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6. Database Design Overview

6.1 User Model

Purpose:

Store user identity, authentication, profile, session, and access information.

Major Information Stored:

- Aadhaar Number
- Role
- Profile Information
- Email Address
- Phone Number
- Session Information
- Login Tracking Information
- Location Restrictions
- Application References

6.2 Service Model

Purpose:

Store service applications and workflow information.

Major Information Stored:

- Applicant Information
- Service Type
- Status
- Form Data
- Review Information
- Approval Information
- Rejection Information
- Timestamps

Supported States:

- Pending
- Reviewed
- Approved
- Rejected

6.3 Area Model

Purpose:

Store Panchayat-level administrative information.

Major Information Stored:

- Panchayat Name
- State
- Geographic Information
- Office Information
- Service Availability

6.4 Polygon Model

Purpose:

Store geographic boundary information used for location validation.

Major Information Stored:

- State Coordinates
- Boundary Data
- Location Validation Data

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7. Future Scope

The following enhancements are planned for future versions:

Service Enhancements

- Custom Service Creation
- Dynamic Form Builder
- Service Search System

Document Enhancements

- PDF Generation
- DOCX Generation
- Downloadable Certificates
- Digital Signatures

Workflow Enhancements

- Notifications

- Email Updates
- SMS Updates
- Escalation Workflows

Administrative Enhancements

- Analytics Dashboard
- Approval Reports
- Audit Reports
- Activity Monitoring

Security Enhancements

- Multi-Factor Authentication
- Advanced Audit Logging
- Security Monitoring Dashboard
- Profile images currently use demonstration avatars stored locally within the application. User-uploaded profile images and cloud-based media storage are planned for future releases.
- **Cloudflare** *(Planned / Future Integration)* — Intended for DNS management, performance optimization, SSL enhancement, caching, and protection against automated bots and malicious traffic.

User Experience Enhancements

- Multi-language Support
- Progressive Web Application
- Offline Functionality
- Enhanced Accessibility

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8. Conclusion

LokVeda demonstrates the practical implementation of a secure digital governance platform tailored for Gram Panchayat operations.

The system combines:

- Authentication
- Geographic Verification

- Workflow Management
- Role-Based Authorization
- Service Administration

within a unified platform designed to remain accessible to non-technical users.

The project successfully transforms traditional administrative workflows into a structured digital system while maintaining usability, accountability, and security.

Its modular architecture, workflow design, and security mechanisms provide a strong foundation for future development and real-world deployment.